

2024

PHYSICS — HONOURS — PRACTICAL

Paper : SEC-1P

(Introduction to Graph Plotting and Programming)

Full Marks : 100

The figures in the margin indicate full marks.

DAY - 2

LNB – 20, VIVA – 20, Programming – 60

Answer any one question from each group.

Group - A

1. Save the following data table in a file :

x	0.5	0.65	0.8	0.95	1.1	1.25	1.4	1.55	1.7	1.85	2.0
$f(x)$	0.297	0.688	1.338	2.319	3.706	5.58	8.019	11.106	14.926	19.563	25.107

Fit the data points of the second row against that of the first row using GNU PLOT with the equation $f(x) = x^a b^x$. Write the values of the fitted parameters a , b . Show the fitted curve along with the data points in the same plot. Set the title, x-label, y-label as 'Data fitting', ' x ' and ' $f(x)$ ' respectively. Save the plot in suitable format. 20

2. Plot $2x, x^2, \frac{x^3}{4}$ on same graph. The line width of the three curves to be chosen to be different.

Set the range of x values from -2 to $+2$ and the range of y values from -4 to $+4$. Show both the axes on the plot. Set title 'Plot of different powers of x '. The label of x and y axis to be ' x values' and ' y values' respectively. Store the plot in a file and print it. 20

3. Plot the following polar functions for $0 \leq \theta \leq 2\pi$:

$$r(\theta) = 20 \sin(2\theta) \quad \text{and} \quad r(\theta) = 20 \cos(2\theta)$$

Set different line styles and line widths for these two plots. Put the title, x-label and y-label as 'Polar plots', ' θ ' and ' $r(\theta)$ ' respectively. Set sample 800. Save the plot in suitable format. 20

4. Plot the following function in the range $0 \leq x \leq 2$

$$f(x) = \frac{ax^3}{e^{bx} - 1}$$

for $a = 1000$ and $b = 8, 9, 10$. Put different line styles and line widths for different b values. Set title, x-label and y-label 'Black body radiation', 'x' and 'f(x)' respectively. Set samples to 800. Save the plot in suitable format. 20

5. Plot the following parametric functions with $0 \leq t \leq 2\pi$:

$$x(t) = [1 + \cos t] \cos t, \quad y(t) = [1 + \cos t] \sin t$$

Put the title as 'Parametric plot'. Set samples to 800. Save the plot in suitable format. 20

6. Make the following data table in a file.

x	3.40	9.56	11.45	13.89	17.5
y	8.04	7.05	6.75	6.38	5.82

Plot the data in second column against that of the first column. Set the range of x values from 0 to 20. Try to fit the data with the equation of type $f(x) = mx + c$. Show the points and the fitted line on the same plot. Mention the values of m and c after fitting the data. Store the plot in a file and print it. 20

7. Define the following function $f(x) = \begin{cases} x \sin x & x < 0 \\ x + \sin x & 0 \leq x \leq \frac{\pi}{2} \\ \cos x + x + 1 & x > \frac{\pi}{2} \end{cases}$

Set the range of x values from $-\pi$ to π and the range of y values from $-\pi$ to $+2\pi$. Choose sample size suitably to make the function appear continuous. Show both the axes on the plot. Set title 'Continuous function without derivative at all points'. Plot the function, store it in a file and print it. 20

(3)

B(1st Sm.)-Physics-H/Pr./SEC-1P/Common Inst./
CCF (Day-2)

Group - B

8. (a) Using Newton-Raphson method with starting point $x_0 = -4$, determine the root of the following equation correct up to three decimal places :

$$x^2 \sin x - e^x \cos x = 0$$

Print the root.

30

- (b) Find the non-trivial factors of the integer 2310. Print all the factors.

10

9. (a) Solve for the real root of the equation $x^3 + x^2 + x + 7 = 0$ using Bisection method correct upto 3 decimal places (Root lies between -3 and 0). Print the root.

30

- (b) Create a list A of first 50 natural numbers. From this list, select the even and odd numbers in two separate lists B and C . Print all the three list.

10

10. (a) A first-order differential equation is given as below :

$$e^{-y} \frac{dy}{dx} = \sin x \text{ with } y(0) = 0$$

Create a list (say X) with 100 x -values between $[0, 1]$. Write a Python program to solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = -\ln \cos x$. Compare the calculated and exact values at $x = 1$. Print both the values.

30

- (b) Two vectors are given as $A = -2i + j + 5k$ and $B = 3i - 2j - k$. Calculate and print $A.B$.

10

11. (a) Using Newton-Raphson method with starting point $x_0 = -4$, determine the root of the following differential equation correct up to three decimal places :

$$e^{-2x} \sin x + \cos x = 0$$

Print the root.

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- (b) Zeta function is defined as

$$\zeta(n) = \sum_{k=1}^{\infty} \frac{1}{k^n}$$

Calculate $\zeta(4)$ correct upto three decimal places.

10

12. (a) A first-order differential equation is given as below :

$$\frac{dy}{dx} = y - x \text{ with } y(0) = 2$$

Create a list (say X) with 100 x -values between $[0, 1]$. Write a Python program to solve this differential equation using Euler method for x -values in the list X . It is given that the exact solution is $y(x) = e^x + x + 1$. Compare the calculated and exact values at $x = 1$. Print both the values. 30

- (b) Write a Python program to calculate the roots of the following quadratic equation :

$$x^2 + 5x + 6 = 0.$$

Print the roots.

10

13. (a) Write a python program to find the root of the transcendental equation correct up to three decimal places

$$\ln(2x) \sin(3x - 1) = 0$$

in between $[2, 3]$ using Bisection method. Print the root.

30

- (b) Write a Python program to calculate the sum (S) of the following series up to 5 terms :

$$S = \frac{1}{2^2} + \frac{2}{3^2} + \dots + \frac{5}{6^2}$$

10

14. (a) Write a Python program to perform the following matrix operation :

$$C = 3A^T + 4B$$

for matrices

$$A = \begin{pmatrix} 1 & 0 & -3 \\ -2 & 4 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & -4 \\ 3 & -1 \end{pmatrix}$$

Print C .

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- (b) Write a program to store first n odd integers in a list. Find the sum and average of the numbers in the list and store them in a tuple.

Print the list and the tuple. Take $n = 20$ as runtime input.

10